

Education

McGill University & Mila - Quebec AI Institute - Montreal, Canada PhD in Computer Science, supervised by Prof. David Rolnick	Sep 2021 - May 2026
McGill University - Montreal, Canada BSc in Physics, thesis supervised by Prof. Natalya Gomez	Sep 2014 - May 2017

Experience

Research Scientist, Allen Institute for AI (AI2) Engineering Intern Researching applications of machine learning for remote sensing on the Earth Systems team	September 2024 - Present June 2024 - August 2024
Machine Learning Engineer, NASA Harvest Researched machine learning for remote sensing and agriculture, supervised by Prof. Hannah Kerner	Apr 2020 - August 2025
Machine Learning Research Intern, Small Robot Company Researched semi-supervised object detection for agriculture, supervised by Dr. Thomas Walters	Summer 2022
Machine Learning Engineer, Okra Solar Developed data and machine learning pipelines for off-grid solar energy systems	Sep 2019 - Aug 2021
Researcher, European Center for Medium Range Weather Forecasting Researched applications of machine learning applied to drought prediction in Kenya	Summer 2019
Machine Learning Engineer, Drop Technologies Developed and deployed end to end machine learning pipelines for Fortune 500 clients in various domains	Oct 2017 - May 2019

Selected Publications

A full list of publications is available on Google Scholar

OlmoEarth: Stable Latent Image Modeling for Multimodal Earth Observation

Henry Herzog, Favyen Bastani, Yawen Zhang, **Gabriel Tseng**, Joseph Redmon and others
CVPR 2026 (link)

Galileo: Learning Global & Local Features of Many Remote Sensing Modalities

Gabriel Tseng, Anthony Fuller, Marlena Reil, Henry Herzog, Patrick Beukema, Favyen Bastani, James R. Green, Evan Shelhamer, Hannah Kerner, David Rolnick
ICML 2025 (link)

Lightweight, Pre-trained Transformers for Remote Sensing Timeseries

Gabriel Tseng, Ruben Cartuyvels, Ivan Zvonkov, Mirali Purohit, David Rolnick, Hannah Kerner
Climate Change and AI workshop at NeurIPS 2023 (link)

Awards and Grants

Leadership in Environmental And Digital innovation for Sustainability (LEADS)	2021, 2022, 2023
Satellite-Enabled Food Security Dashboard to Close Critical Data Gaps in	2022
Highly Food Insecure Maui County (NASA Environmental Equity and Justice Grant)	
European Center for Medium Range Weather Forecasting Summer of Weather Code	2019
James McGill scholarship	2014

Service

Machine Learning for Remote Sensing Workshop @ ICLR - Organizer	2023, 2024, 2025, 2026
NeurIPS datasets and benchmarks track - Reviewer	2022, 2023

Additional Interests

Endurance sports (ran 6 ultra-marathons & a DIY-Ironman), rock climbing, SCUBA diving (advanced open water diver)
Languages: English (fluent), French (fluent), Mandarin (basic)